AMRITA VISHWA VIDYAPEETHAM | SCHOOL OF COMPUTING CHENNAI

QUANTUMSAFE VPN

Post-Quantum Cryptography Secured Android VPN

HERE'S HOW TO USE IT:

1. Open App & Configure Server IP
2. Authenticate Server with ML-DSA-65
3. Ephemeral KEM Handshake (ML-KEM-1024)
4. One-Tap Encrypted Post-Quantum Tunnel

AUTHORS

P. Abishek Singh (U4CYS23001)
R. S. Sanjaay Krishna (U4CYS23043)
Supervisor: Dr. G. Saranya, Asst. Prof.

GITHUB REPO

Project Overview

A snapshot of the what, why, and how.

QuantumSafe VPN is an end-to-end post-quantum cryptographic VPN engineered to protect mobile IP communications against Harvest Now, Decrypt Later (HNDL) attacks and future quantum computers running Shor's algorithm. By replacing classical RSA and ECC with finalized NIST standards (FIPS 203 ML-KEM-1024 and FIPS 204 ML-DSA-65), it delivers authenticated, hardware-accelerated AES-256-GCM encrypted tunneling between an unprivileged Android client and a hardened Linux security gateway with automated dynamic IP

Goals & KPIs

What are we aiming for?

Immunizes mobile communications against quantum cryptanalysis using NIST standards.

Zero-leak full tunneling with sub-40ms PQC handshake & real-time telemetry.

Architecture Overview

A visual mind map illustrating the architecture.

KEY FEATURES

- ML-KEM-1024 (FIPS 203) Key Encapsulation
- ML-DSA-65 (FIPS 204) Gateway Signature
- Hardware AES-256-GCM AEAD Tunneling
- HKDF-SHA256 (RFC 5869) Key Derivation
- 64-bit Anti-Replay Sliding Window

GATEWAY

- Linux Kernel Virtual TUN Interface (tun0)
- Dynamic IPv4 Lease Subnet (10.8.0.0/24)
- Kernel iptables NAT Masquerading
- Length-Prefixed TCP Stream on Port 7000
- High-Throughput Async Non-Blocking I/O

QUANTUMSAFE VPN

ANDROID CLIENT

- Native VpnService IP Packet Capture
- Pure-Kotlin Keccak & Lattice Engine
- Kernel Socket protect() Loop Isolation
- Jetpack MVVM & Material 3 UI
- Live Cryptographic Telemetry Dashboard

TECH STACK

- Backend: Python 3.11, Linux TUN, iptables
- Client: Kotlin, Android SDK 34, Coroutines
- Crypto: pure-pqc, BouncyCastle, cryptography
- Testing: Wireshark, Scapy, pytest, LMK tests
- OS: Kali Linux 2024.2 & Android 14

Impact & Conclusion

Highlights the value and outcome of the project.

QuantumSafe VPN has a strong impact by making quantum-safe mobile networking robust, practical, and highly accessible. Its integration of lattice-based post-quantum cryptography (NIST FIPS 203 & 204) with native Android VpnService proves that smartphones can defend against Harvest Now, Decrypt Later (HNDL) attacks without sacrificing battery life, packet throughput, or user experience. The project sets a solid foundation for future advancements in post-quantum mobile cybersecurity.

Insights & Results

what the platform detects and delivers.

This project provides deep empirical insights into post-quantum mobile tunneling. Rigorous verification across cryptographic units, wire protocols, and physical testbeds confirmed 100% pass rates across all NIST KAT vectors. Active adversary injection verified zero packet leakage, 100% rejection of bit-flipping attacks via Galois MAC tags, and sub-40ms PQC handshakes with sustained 50+ Mbps throughput on Google Pixel (Android 14).